

AI-based Solution for Learning Disabilities

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Student Declaration

We hereby declare that the work presented in the report entitled ”**AI-based Solution for Learning Disabilities**” submitted by us for the partial fulfilment of the requirements for the degree of B.Tech. at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under the guidance of **Prof. Md. Shad Akhtar** , **Co-advisor: Mr. Aditya Bhardwaj**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Aarushi Verma & Aayush Gupta

Place & Date: IIIT Delhi, April 21, 2026

Certificate

This is to certify that the above statement made by the candidates is correct to the best of our knowledge.

Prof. Md. Shad Akhtar
Co-advisor: Mr. Aditya Bhardwaj

Place & Date: IIIT Delhi, April 21, 2026

Abstract

Dyslexia affects a significant number of students in India, yet nearly all learning-support technologies available today have been built around Western, predominantly English-language datasets. This mismatch makes them largely unsuitable for the multilingual, phonetically diverse environment that Indian learners actually inhabit. In the first semester of this project, we designed and deployed a secure, web-based multimodal data collection platform capable of gathering reading, writing, speech, typing, mathematical comprehension, and logical aptitude responses from school-age children, wrapped inside a role-based access framework covering schools, parents, students, and administrators.

This semester extended that foundation in several significant directions. The platform was migrated from a local university server to a production-grade cloud environment on Amazon Web Services, comprising an EC2 compute instance, an RDS-hosted MySQL database, and an S3 bucket for media assets. A custom domain (dyslexia.flamenlp.in) was acquired and configured with HTTPS through a valid SSL certificate, making the system publicly accessible and secure. Consent and legal compliance were strengthened through a legally reviewed Terms of Service, Privacy Policy, and parental consent form. The task library was substantially expanded: each class now receives between six and twelve NCERT textbooks aligned tasks for that grade level, verified against Bloom’s Taxonomy by domain specialists. Two new task categories were introduced, Listening Comprehension and Visual Aptitude (under Aptitude tasks), broadening the cognitive signals captured. Camera access (with explicit per-student consent) was added to record synchronized audio-video during task attempts, providing richer behavioral data for future model training. A post-task feedback and star-rating system was integrated to log student experience. Performance statistics (typing speed in characters per second, verbal interaction pace from audio analysis, scores and more) are now computed and displayed to students and parents in real time. An Android APK was built to extend platform reach to mobile devices.

Together, these changes bring the platform to a state of full production readiness, publicly hosted, legally compliant, and equipped with a richer multimodal dataset pipeline than was possible at the end of Semester 1.

Keywords: Dyslexia, Educational Technology, Multimodal Dataset, AWS Deployment, NCERT Tasks, Bloom’s Taxonomy, Indian Languages, Accessibility, Gamification, Audio-Video Recording.

Acknowledgment

We would like to express our heartfelt gratitude to Prof. Md Shad Akhtar and our Phd guide Aditya Bhardwaj for their consistent guidance, feedback, and domain expertise, which shaped this project's direction. We also thank IIIT-Delhi for providing research infrastructure, and the participating schools, parents, and students who enabled real-world deployment. This work would not have been possible without their cooperation.

Work Distribution

The work completed during this semester focused on the design, development, and deployment of the dyslexia screening platform, encompassing task engineering, multi-modal data collection, cloud infrastructure, and accessibility features. The project was carried out collaboratively, with both members contributing to core system architecture, database design, testing, and iterative refinement based on pilot-school feedback. The contribution of each member is summarized below:

Aarushi Verma

- **Camera & Microphone Integration:** Implemented browser-native A/V capture with a three-choice consent flow (video and audio/audio-only/decline), a live camera preview widget, and independent per-stream toggle controls.
- **Audio comprehension and Visual comprehension Task Added:** Integrated the Listening Comprehension and Visual Aptitude task types into the student UI, audio playback widget with per-question response timing, pattern-display renderer, and unified save-submit-retake flow consistent with all other task types.
- **Task Library Population:** Expanded per-class task sets (based on Bloom's taxonomy) from 6 tasks per class to 12 per class into the student interface; verified correct dynamic assignment of tasks by class level.
- **Student Platform and Task Interface Enhancement:** Developed the interactive Student Dashboard and task interfaces, integrating frontend logic for gamification (badges, scoring), task navigation, and real-time state management. Enhanced the UI and workflow to fit the target users.
- **Parent Platform Improved:** Developed the parent portal for consent management, child account creation, and longitudinal progress visualization; extended this semester with a dedicated profile page for account and consent-status review.
- **Accessibility Suite:** Architected the custom DOM manipulation engine for the accessibility toolbar, creating toggle features like the interactive reading ruler, OpenDyslexic font toggles, and high-contrast monochrome modes.

Aayush Gupta

- **AWS Cloud Deployment:** Migrated the platform from the university server to AWS, Flask on EC2, MySQL on RDS (automated backups, managed patching), and all media assets in a private S3 bucket, achieving clean separation of compute, database, and storage.
- **Domain, Subdomain & HTTPS/SSL:** Acquired flamenlp.in, configured dyslexia.flamenlp.in with a DNS A record, provisioned an SSL/TLS certificate, and enforced HTTP-to-HTTPS redirection platform-wide, satisfying the security requirements for collecting data from minors.

- **Feedback Data Pipeline:** Designed the feedback system (star ratings and free-text comments linked to attempt records), implemented the API endpoints to receive and store ratings, and surfaced aggregated feedback data in the Admin Dashboard for task quality monitoring.
- **Admin Platform Improved:** Built the centralized Admin Panel to monitor global system health, aggregating data to visualize cumulative statistics and participation trends across all partner schools.
- **School Platform Enhancement:** Developed the institutional School Dashboard, enabling granular class-wise management, bulk student/parent onboarding, and classroom-level performance tracking.
- **API Architecture:** Architected the robust backend infrastructure using Flask and Python, designing custom API endpoints to handle complex role-based access control and high-concurrency data traffic.

Joint Work

- **Database Schema Evolution:** Collaboratively structured the relational database to support the interconnected hierarchical relationships (School → Class → Student → Parent) while ensuring data anonymity. Extended it to support new modalities, per-attempt media-type flags, feedback records, and difficulty metadata, while preserving full backward compatibility.
- **Task Design and Bloom's Taxonomy:** Curated tasks for each category (Reading, Writing, Reading Comprehension, Mathematical Comprehension, Aptitude, Listening) aligned with NCERT-style pedagogical frameworks. Jointly classified each task by Bloom's cognitive-demand level (recall through evaluation), with formal documentation reviewed by domain specialists and informed by identified literature gaps.
- **Progress Statistics:** Jointly implemented the performance statistics and badges display pipeline that surfaces performance metrics (typing speed in chars/sec, verbal pace in words/min, accuracy, score and more) to students on a post-task statistics screen.
- **Android APK:** Packaged the production web platform as a WebView-based Android APK; tested across three device models for task functionality, camera/microphone access, and full UI correctness; made available for sideload from the platform website.
- **UI/UX Design Enhancement:** Led the design of the responsive interface, ensuring an intuitive user journey for non-technical parents and students.
- **Demo Video & Onboarding Guide:** Jointly produced a walkthrough video and annotated step-by-step written guide covering the end-to-end workflow (school setup → parent registration → student task flow); published on the platform home page to reduce onboarding friction for non-technical users.
- **Testing:** Conducted systematic cross-browser and cross-device testing (Chrome, Firefox, Android) covering task flows, A/V recording, consent gating, and S3 media retrieval; iterated on bug fixes and incorporated feedback from pilot-school testers.

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Chapter 1

Introduction

1.1 Background

Dyslexia is a neurodevelopmental learning disorder that affects reading fluency, written expression, and phonological awareness. It is one of the most prevalent Specific Learning Disabilities (SLDs) worldwide, yet its detection and management in India remain significantly underserved. The majority of automated screening and support tools in use today were trained on data from Western, English-dominant populations, and they reflect reading behaviors, error patterns, and phonetic systems that differ substantially from those encountered in Indian classrooms.

Indian learners typically navigate environments where English is a second or even third language, where classroom instruction may switch between scripts, and where family literacy levels vary widely. These contextual factors produce dyslexic error patterns that current tools are simply not designed to detect. As a result, most Indian students with dyslexia go unidentified, and those who are identified lack access to culturally appropriate assistive technology.

1.1.1 Motivation

The absence of an Indian-specific multimodal dyslexia dataset is the root cause of this gap. Without such data, it is not possible to train models that understand Indian reading behavior. Addressing this data scarcity problem was the founding motivation for this project.

Previous semester, we established that passive data collection, asking students to simply submit responses, is unlikely to scale in a school setting without strong incentives. We therefore designed the platform around a reciprocal "Give and Take" model: students and parents receive genuinely useful feedback (scores, statistics, progress tracking) in exchange for contributing behavioral data. This semester we built on this model by making the platform publicly accessible on the web, legally sound, and significantly richer in the types of data it collects.

1.1.2 Problem Statement

There is no existing, ethically collected, multimodal dyslexia dataset that represents Indian school children across age groups and task types. Without such a dataset, it is not feasible to train or validate AI screening models appropriate for the Indian educational context. The platform described in this report directly addresses this problem by providing the verified data-collection infrastructure needed to produce that dataset at scale.

1.1.3 Objective

The objectives of this project, carried across both semesters, are:

- To build and deploy a secure, publicly accessible, multimodal data collection platform for Indian students.
- To collect reading, writing, speech, typing, mathematical comprehension, listening comprehension, visual aptitude, and logical reasoning responses.
- To integrate accessibility features and gamification to reduce anxiety and sustain engagement.
- To ensure full ethical compliance including parental consent, data anonymisation, and legal documentation.
- To establish the infrastructure for training future AI models tailored to Indian dyslexic learners.

The current phase encompasses platform engineering, cloud deployment, domain and SSL configuration, legal compliance finalization, task library expansion, and mobile accessibility. Machine learning model development will proceed in the subsequent phase once sufficient data has been collected.

Chapter 2

Literature Review

Research in dyslexia detection has increasingly moved away from single-modality screening (such as paper-based reading tests) toward behavioral, multimodal approaches that capture a wider range of cognitive signals. Studies have shown that speech error patterns, orthographic reversals, reading pace and hesitation, keystroke dynamics, and comprehension timing all provide meaningful diagnostic information that improves screening accuracy over text-only assessments [3] [6].

A 2025 scoping review by Yap, Aruthanan, and Chin surveyed the landscape of AI-driven dyslexia research and concluded that while the field is moving toward early detection and personalized intervention using diverse data points, there remains a substantial gap in tools designed for non-English learners [4]. Similarly, Zhao et al. demonstrated that AI-assisted reading tools can meaningfully improve literacy outcomes, but their effectiveness depends on the availability of high-quality, representative training data [5]. Neither of these contributions addresses the Indian context specifically, reinforcing the need for the kind of dataset infrastructure we are building.

The role of mobile and web-based platforms in literacy support has been examined by Molina Villamarín and Mena Vargas, who found that while digital strategies improve engagement, they often lack integration with formal school structures and consent-based participation workflows [1]. Our platform addresses this gap explicitly through its school-mediated onboarding process and IRB-aligned parental consent chain.

Piferi et al. proposed a multimodal framework for integrating Large Language Models into therapeutic education for children with dyslexia, noting the importance of combining speech, text, and interaction-level signals in a single pipeline [2]. This work informed our decision to record synchronized audio-video during task attempts rather than treating audio and video as separate streams.

On the technical side, the use of Amazon Web Services for hosting educational platforms at scale is well-established in the literature on EdTech infrastructure. The combination of EC2 for compute, RDS for relational data, and S3 for media storage provides the separation of concerns and independent scalability that is essential when audio and video files grow rapidly with user participation. Prior work on NCERT-based curriculum design confirms the appropriateness of using official Indian textbooks as the source of reading and comprehension passages, as they reflect the standardized

vocabulary and phonetic exposure typical of Indian students by grade level.

In summary, existing literature establishes the scientific value of multimodal dyslexia data but provides no ready-made solution for the Indian educational context. Our work occupies this gap by combining a verified school workflow, legally sound consent mechanisms, NCERT aligned task content, and a cloud-hosted platform capable of operating at a large scale.

Chapter 3

Methodology

We focused on transforming a locally hosted, technically functional prototype into a production-grade, publicly accessible platform with a highly efficient UI and user flow. This involved **cloud migration, domain and certificate configuration, legal compliance finalization, substantial task library expansion, progress statistics enhanced user flow and a set of new data-capture capabilities**. The core architecture from Semester 1 was retained and extended rather than replaced.

3.1 System Architecture Overview Recap

For completeness, the platform follows a four-layer architecture: a Flask and Python backend handling routing, REST-like API endpoints, and role-based access control; an HTML/CSS/JavaScript frontend with Jinja2 templates; a MySQL storage layer for structured data; and a file storage layer for audio and video assets. Four user roles are supported :School, Parent, Student, and Admin, each with dedicated dashboards and access controls. The six core tasks (Reading Aloud, Writing, Typing, Reading Comprehension, Mathematical Comprehension, and Aptitude) were fully operational at the close of Semester 1. This report focuses on the additions and extensions made during Semester 2.

Role	Key Features
School	Class creation, bulk student listing (CSV or manual), linking students to parents, enabling participation, viewing school-wide statistics and task completion heatmaps
Parent	Independent signup, child registration, demographic submission, consent authorization, viewing detailed progress and task scores, launching student portal

Continued on next page

Role	Key Features
Student	Accesses student portal via parent, performs multimodal tasks, saves progress, submits attempts, retakes tasks, views badges, scores, and analytics
Admin	Monitors platform-wide statistics, manages schools, views global participation analytics, audits data quality, and oversees system health

Table 3.1: System roles and associated privileges. The platform supports role-based workflows to ensure ethical, secure, and validated multimodal data collection.

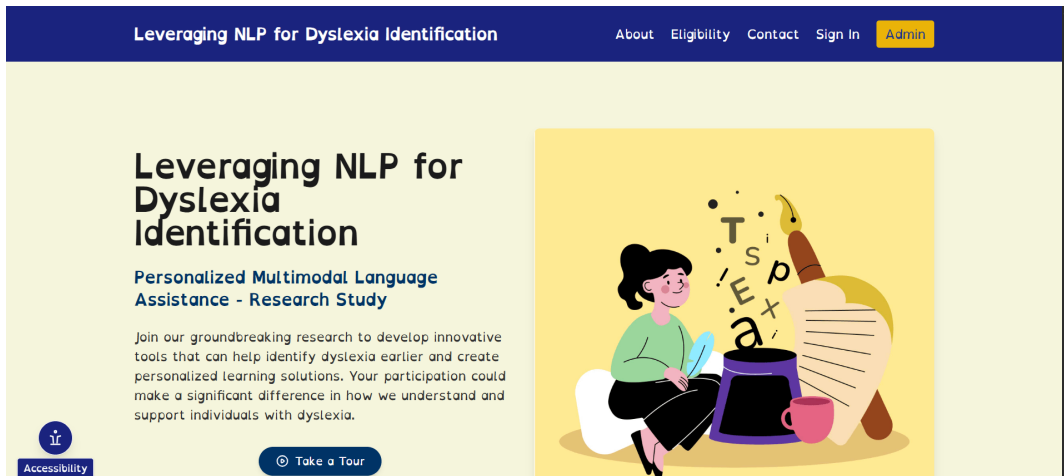


Figure 3.1: Main landing page

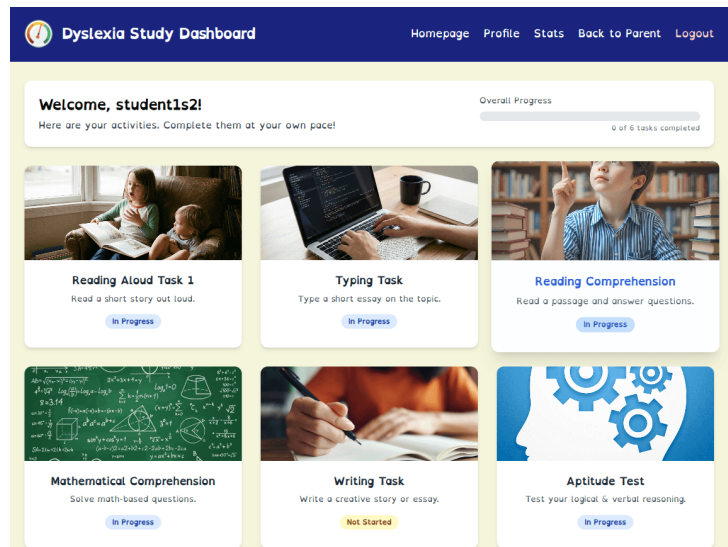


Figure 3.2: Student Dashboard tasks

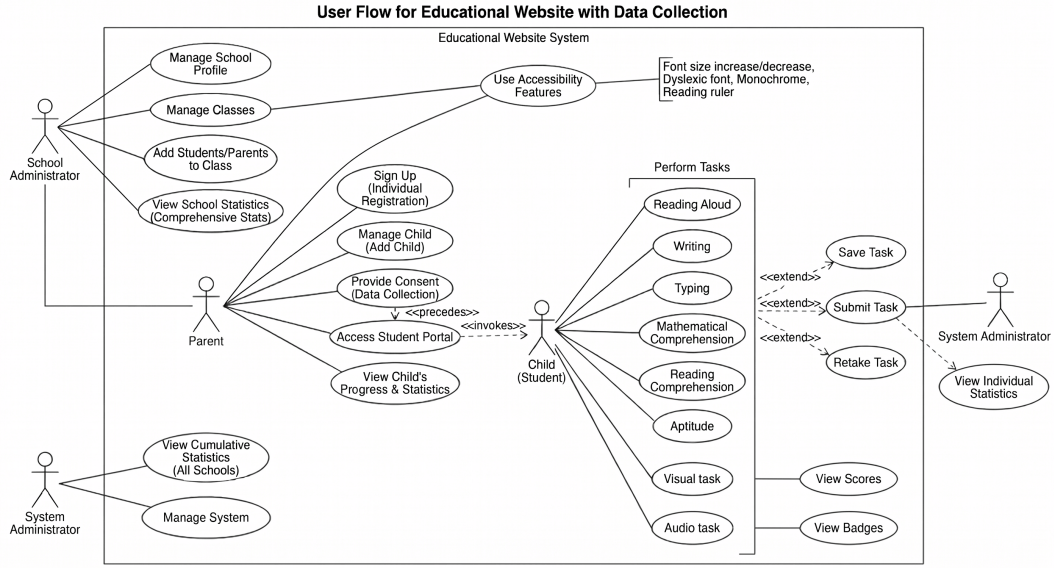


Figure 3.3: High-level user flow illustrating the interactions between School, Parent, Student, and Admin workflows along with backend and database components.

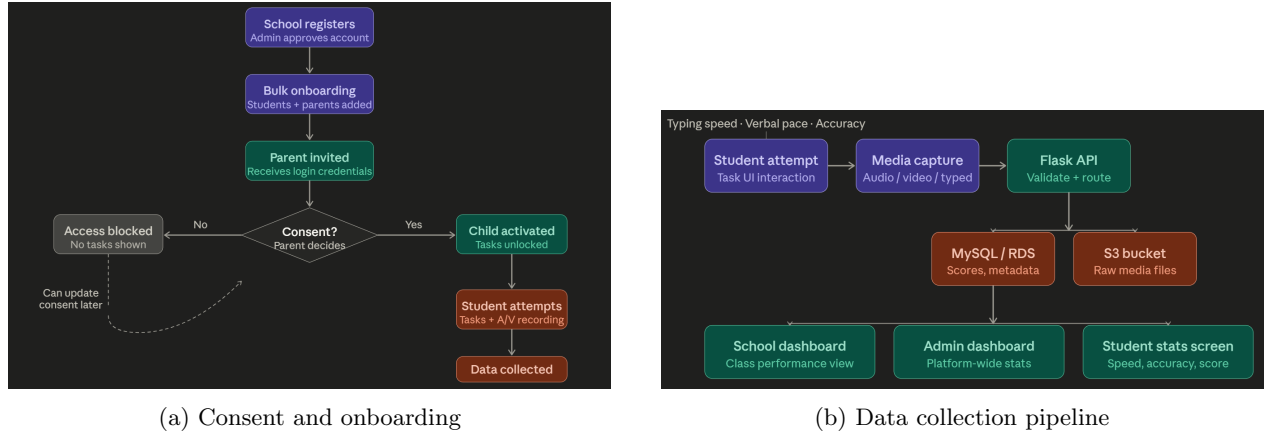


Figure 3.4: End-to-end workflow from onboarding to task execution and data collection

- **Database Design** : The MySQL schema is normalized, hierarchical, and engineered to ensure data integrity, provenance, and scalability during large-scale multimodal data collection. The database is organized into entity tables, task-definition tables, attempt provenance tables, and task-specific progress tables. The schema incorporates Foreign Key Constraints, Unique Constraints, Attempt Logic, File Integrity, PII and Task Data separation mechanisms to ensure correctness and consistency.

Task	Data Collected	Upgradations
Reading Aloud	Audio via Web Audio API, pronunciation cues, reading pace, pause durations	Audio+video recording, Task database increased to 12 per class, Audio analysis based statistics displayed post-task
Writing	Hand-written images of the given task	Audio+video recording, Task database increased to 12 per class
Typing	Millisecond-level keystroke timing, speed, reversal patterns, latency bursts	Audio+video recording, Task database increased to 12 per class, Typing strokes and speed statistics displayed post-task
Reading Comprehension	MCQ responses, response time per question	Audio+video recording, Task database increased to 12 per class
Mathematical Comprehension	Error propagation patterns, numeric mistakes, reasoning flow	Audio+video recording, Task database increased to 12 per class
Aptitude	Cognitive strategies, logical reasoning accuracy, question-level timing	Audio+video recording, Task database increased to 12 per class
Listening Comprehension (Aptitude)	Audio processing, MCQ responses	Newly added
Visual Aptitude (Aptitude)	Pattern recognition, MCQ responses	Newly added

Table 3.2: Tasks and multimodal data collected.

To preserve data integrity, typing tasks implement copy-paste prevention at the browser level, ensuring that submitted text reflects the student’s own composition rather than pasted content.

3.3 Cloud Deployment on AWS

One of the most significant engineering milestones of this semester was the complete migration of the platform from a university server to Amazon Web Services. The deployment uses three AWS services in combination:

- **EC2 (Elastic Compute Cloud):** The Flask application server runs on an EC2 instance configured with the appropriate security groups to allow HTTP/HTTPS traffic. The instance

was provisioned, the application environment was set up, and the server was configured to start the Flask application as a background service on reboot.

- **RDS (Relational Database Service):** The MySQL database was migrated from local storage to an AWS RDS instance. RDS provides automated backups, managed patching, and multi-AZ failover capability, significantly improving the data reliability compared to the previous setup. The existing schema required no structural changes for the migration.
- **S3 (Simple Storage Service):** All media files , audio recordings, video recordings, and uploaded writing samples , are now stored in a private S3 bucket. Files are uploaded directly from the backend with randomized keys and are referenced in the database by their S3 path. This decouples media storage from compute, allowing the EC2 instance to remain stateless.

This architecture follows standard cloud best practices: compute, database, and object storage are handled by separate, independently scalable services. The separation is important given that media files are expected to grow rapidly once school deployments begin, and it would be impractical to store audio and video on the compute instance itself.

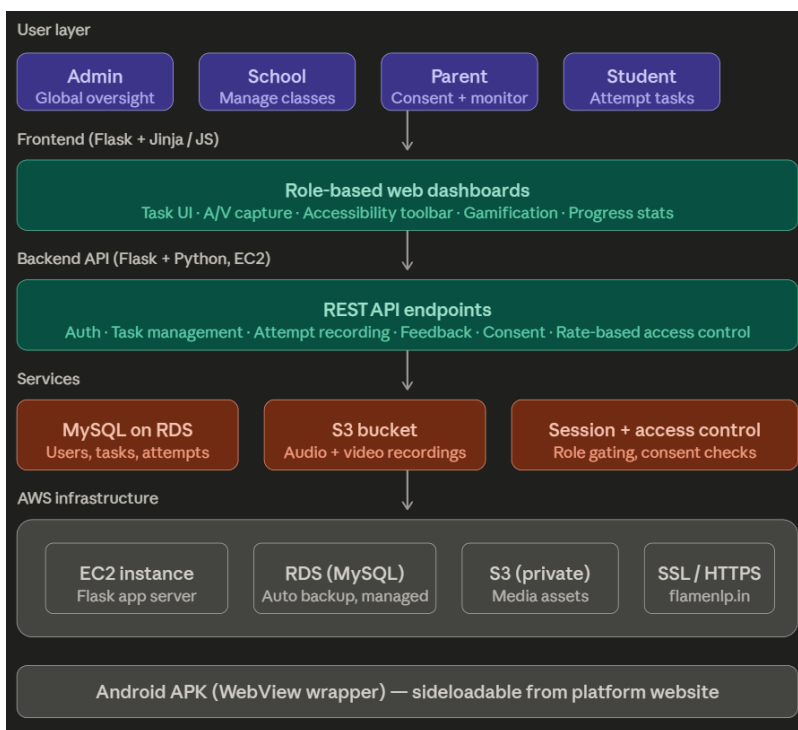


Figure 3.6: System architecture and data pipeline illustrating the the full stack from user roles down to AWS infrastructure.

3.4 Domain Purchase, Subdomain Configuration, and HTTPS

To make the platform publicly accessible under a stable, professional address, a domain was purchased from GoDaddy under the name flamenlp.in, aligned with the FLAME NLP lab at IIIT Delhi.

A subdomain, `dyslexia.flamenlp.in`, was created and configured with a DNS A record pointing to the public IP of the EC2 instance. This allows the platform to be reached at a memorable, institution-affiliated URL rather than a raw IP address.

An SSL/TLS certificate was obtained and installed on the server, enabling HTTPS across the entire platform. This is a non-negotiable requirement for any web application that handles data from minors. HTTPS ensures that all data transmitted between the student’s browser and the server, including audio recordings, task responses, and consent form submissions, is encrypted in transit. The certificate was provisioned and the Flask server was configured to redirect all HTTP traffic to HTTPS, ensuring no sensitive data travels over an unencrypted connection.

3.5 Performance Statistics and Gamification

To fulfil the “Give and Take” promise of the platform, that students receive useful feedback in exchange for their data, we implemented a statistics display system that presents computed performance metrics after each task submission.

Reading task analytics. For reading aloud tasks, the submitted audio is transcribed and compared against the ground-truth passage using sequence matching. This produces word-level annotations: correctly read words, skipped words, mispronunciations, and extra insertions. From these, an overall reading accuracy score is computed. In parallel, fluency features are extracted directly from the audio signal: words per minute (WPM) as a measure of reading speed, pause frequency (pauses per minute) derived from silence detection, and a smoothness score that captures the continuity of delivery. Together these metrics give a combined view of reading accuracy, speed, and flow.

Typing task analytics. Typing tasks are evaluated along two dimensions. The final submitted text is compared against the reference passage to compute spelling accuracy and identify character-level error patterns. Simultaneously, the raw keystroke log — capturing every insertion, deletion, and correction in millisecond-resolution — is analysed to compute edit density (the ratio of editing actions to final text length) and sentence structure variability. This dual-channel approach captures both the accuracy of the final output and the effort and revision behavior that produced it, which are independently informative for dyslexia screening.

Score computation. For objective task types (Reading Comprehension, Mathematical Comprehension, Aptitude, Listening Comprehension, Visual Aptitude), scores are computed automatically by comparing student responses against stored correct answers. Time spent per task is recorded and displayed. All scores are normalised using `score/max_score` to enable cross-task comparison.

Student statistics panel. All metrics are surfaced to students and parents through a dedicated statistics panel. The panel displays: task-level scores and time spent, a progress bar showing overall task completion, reading fluency metrics (WPM, pause frequency, smoothness) for reading tasks, and typing behavior metrics (spelling accuracy, edit density) for typing tasks. A cumulative class comparison shows the student’s standing relative to the class average — presented as a single benchmark figure rather than a ranked list, to motivate improvement without demotivating

students through explicit comparison. Personal bests and most-improved indicators are displayed across attempts. Badges are awarded for milestone achievements such as 100% accuracy on a task, consistent participation, and measurable improvement across re-attempts. Parents can view the same statistics panel to monitor their child’s progress longitudinally.

These statistics serve a dual purpose: they provide genuine feedback value that motivates re-engagement, and they add normalized performance features to the dataset that downstream models can use as behavioral predictors.

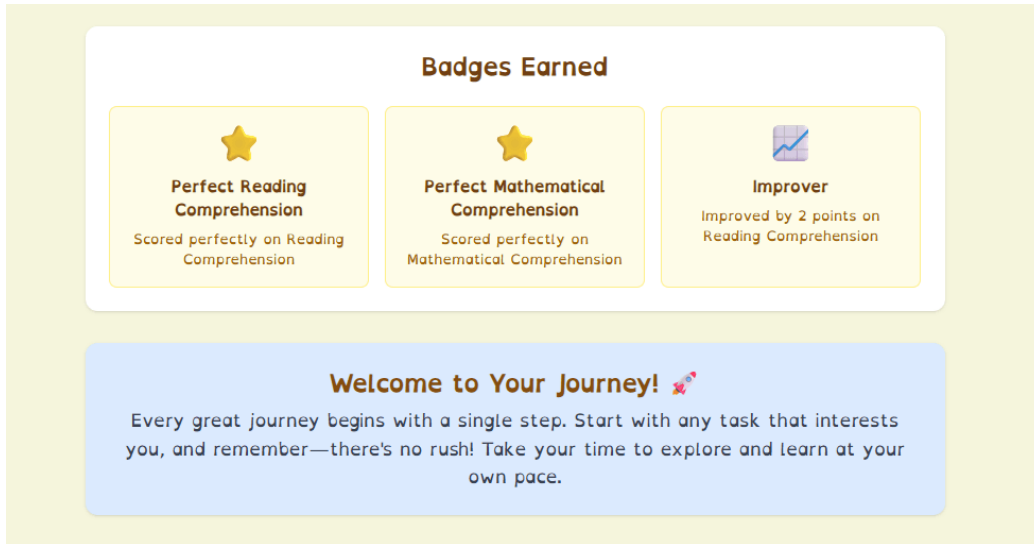


Figure 3.7: Post-task gamification screen showing badges earned by the student

3.6 Camera and Microphone Access with Consent

A significant new data-capture capability introduced this semester is synchronized audio-video recording during task attempts. The motivation is straightforward: audio alone captures pronunciation and reading pace, but video adds facial expression, gaze behavior, posture, and observable signs of reading difficulty that may be valuable for future behavioral analysis models.

The implementation works as follows. When a student begins a task session, the platform prompts for camera and microphone permissions via the browser’s standard media access APIs. The student is shown a clear consent dialog explaining that video and audio will be recorded and stored securely for research purposes. The student has three possible responses: grant both camera and microphone access, grant microphone only (if they prefer not to be recorded on video), or decline both. The task proceeds in all cases, recording permission is not a gate on participation, but the available data stream adjusts accordingly.

When camera access is granted, a live preview of the student’s camera feed is displayed in the top-right corner of the task interface, styled to resemble the self-view familiar from video conferencing tools like Google Meet. This preview serves both as a confirmation that recording is active and as a reassurance to the student that they can see exactly what is being captured. Two toggle buttons, one for video and one for audio, allow the student to pause or resume either stream independently

at any point during the task, giving them ongoing control without interrupting the session. The recorded video and audio streams are combined client-side and the resulting file is uploaded to the S3 bucket upon task submission, referenced in the database with the appropriate attempt ID and timestamp. In cases where only microphone access is granted, an audio-only file is stored instead. The database schema was extended to record which media modalities were captured for each attempt, enabling downstream researchers to filter their analysis by available data types.

3.7 New Task Categories: Listening Comprehension and Visual Aptitude

Two new task categories were added to the platform's cognitive battery this semester.

Listening Comprehension tasks present the student with an audio clip , a spoken passage read at normal pace , followed by multiple-choice questions about its content. The task is designed to capture how effectively a student can process and retain aurally delivered information, which is a dimension of language processing distinct from visual reading. For students with dyslexia, auditory processing difficulties often co-occur with reading difficulties, and capturing this signal allows for a more complete cognitive profile.

Visual Aptitude tasks present the student with visual patterns, sequences, or spatial reasoning problems and ask them to identify the correct continuation or match. These tasks are included because dyslexia sometimes manifests alongside difficulties in visual-spatial processing, and the presence or absence of such difficulties is diagnostically informative. Both task types follow the same save-submit-retake pipeline as the existing tasks and log response timing at the individual question level.

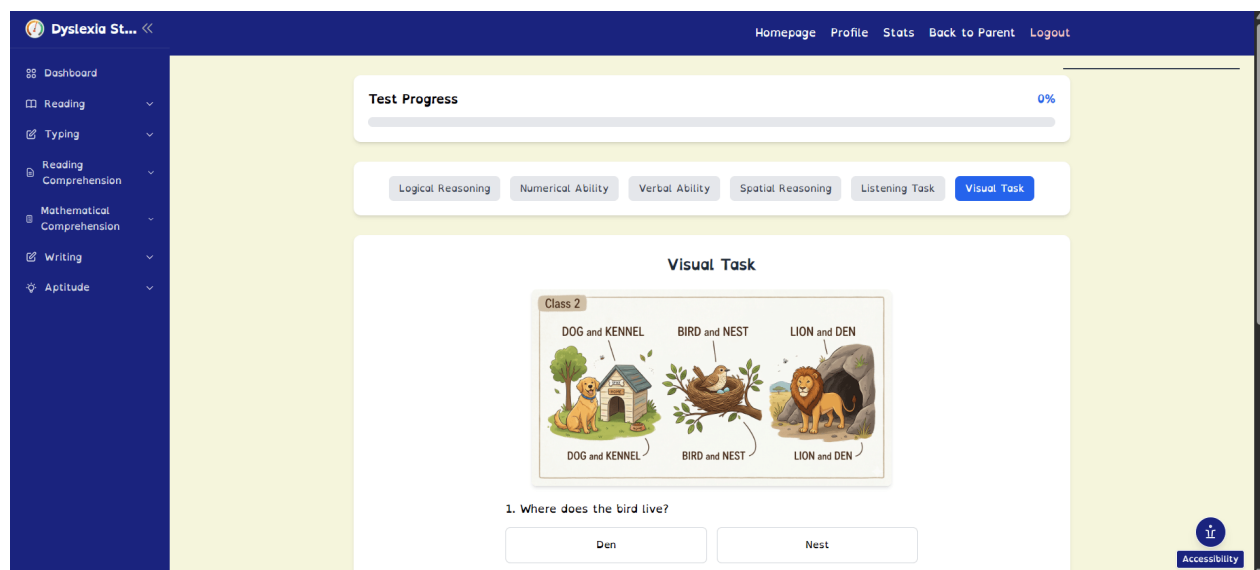


Figure 3.8: Visual task screen on student dashboard(OpenDyslexic font enabled)

3.8 NCERT aligned Task Library Expansion

In Semester 1, each student was assigned a fixed set of six tasks. While this was sufficient to validate the platform, it limited the richness of the dataset. Students who completed all tasks had nothing left to contribute, and the limited diversity of reading passages meant that repeated participation produced diminishing marginal value.

This semester, the task library was substantially expanded. Each class now has between six and twelve tasks, with the exact number depending on grade level and task category. All reading passages, comprehension questions, and sentence prompts were sourced from NCERT textbooks for the corresponding class. Using NCERT content has several advantages: the passages are already calibrated to the reading level expected at that grade, they reflect the vocabulary and sentence structures Indian students encounter in school, and they are in the public domain, removing copyright concerns.

The process of selecting and validating tasks followed the Bloom’s Taxonomy framework. Each task was classified by its cognitive demand level, from basic recall and recognition at the lower end to analysis, evaluation, and creative application at the higher levels. This classification was documented formally and reviewed by domain specialists to ensure the task battery provides adequate coverage across cognitive levels. The resulting documentation serves both as a methodological record for the research community and as a reference for future task additions.

At the backend, tasks are associated with class levels and served dynamically, a student is assigned the task set appropriate for their enrolled class, and the system tracks which tasks they have completed, are in progress, or have not yet started. The expanded task library also introduces a difficulty dimension (Easy, Medium, Hard) at the database level, which is assigned to each task during curation. This information is not currently exposed on the student interface; it is retained as a metadata field for use in model training and will be surfaced in a future release.

3.9 Post-Task Feedback and Rating System

After a student submits any task, they are presented with a brief feedback prompt before being returned to the dashboard. The prompt displays a five-star rating widget and an optional free-text field where the student can describe their experience of the task. Participation in this step is voluntary, the student can skip it entirely, but in practice the low effort involved (a single tap on a star) has resulted in high completion rates during testing.

The ratings and optional comments are stored in a dedicated feedback table linked to the task attempt record. This data serves two purposes. First, it provides a measure of student experience that can identify tasks that are confusing, distressing, or poorly calibrated to the expected grade level. Second, it creates a feedback loop that allows the platform administrators to iteratively improve the task library based on direct student input rather than inference alone.

3.10 Parent Profile Page

In Semester 1, the parent role had a functional dashboard for managing consent and viewing child progress, but there was no dedicated profile page where a parent could review or update their own account information. This semester, a parent profile page was added. It displays the parent’s registered details, lists the children they have registered on the platform, shows the consent status for each child, and provides options to update contact information. The page is accessible from the parent dashboard navigation and follows the same role-gated access control as all other pages.

3.11 Admin: Task Database CSV Export

To support platform maintenance, task auditing, and future scaling, a CSV export feature was implemented in the Admin dashboard. The export compiles the complete task database into a single structured file, including task content (passages, prompts, and questions), associated answer options and correct answers, task category, assigned class level and more. This provides administrators and researchers with a human-readable, portable snapshot of the entire task library without requiring direct database access. The export is generated server-side on demand and downloaded directly from the dashboard, making it practical for sharing with domain specialists for review, for migrating tasks to future platform versions, and for maintaining an offline record of the task corpus as the library continues to grow.

3.12 Demo Guide with Video and Text

User onboarding was identified as a friction point during the pilot study. Parents and school administrators who were unfamiliar with the platform’s workflow found the number of steps , sign up, add a child, obtain school linkage, grant consent, and then launch the student portal ,presented a friction point for first-time users unfamiliar with the multi-step onboarding sequence. To address this, a dedicated Guide page was added to the platform. The page contains a walkthrough video demonstrating the end-to-end workflow from the perspective of a parent and a student, accompanied by a step-by-step written explanation with annotated screenshots. The guide is accessible without login from the home page navigation, making it available to prospective participants before they commit to registering.

3.13 UI Redesign and Responsiveness

This semester involved a substantial redesign of several platform pages beyond the task interface. The landing page and the “Learn More” page were restructured to improve clarity for first-time visitors, particularly parents and school administrators who may be unfamiliar with the research context. Navigation elements were reorganised and footers were fixed consistently across all pages to prevent layout drift on varying screen sizes.

The student dashboard underwent the most significant structural change: tasks are now displayed in a collapsible, category-wise layout rather than a flat list, reducing the visual complexity presented to younger users and making it easier to track progress by task type. Call-to-action buttons were standardised in size, placement, and labeling across all four role dashboards to reduce cognitive friction for non-technical users.

All pages were tested for responsive behaviour across screen widths corresponding to desktops, laptops, tablets, and mobile devices, and layout regressions identified during testing were corrected. The same responsive behaviour was verified through the Android APK on three device models.

3.14 Android APK

To extend access beyond the browser-based interface, the web platform was packaged as an Android APK using a WebView wrapper. The APK allows students to access the platform from Android smartphones and tablets without needing to open a browser, navigate to the URL, and manage bookmarks. The application loads the live production site, meaning it benefits automatically from any updates deployed to the web platform. iOS packaging was explored but was not feasible within the current budget, as Apple’s development ecosystem requires a paid developer account for app distribution. The Android APK is available for direct download from the platform’s website and does not require submission to the Play Store for sideloading.

3.15 Ethical Compliance Updates

The platform prioritizes child safety, data protection, and transparent consent. This semester, the Terms of Service, Privacy Policy, and parental consent form were finalized with the assistance of a legal expert to ensure they meet applicable data protection standards for platforms collecting data from minors. The consent form explicitly describes what data is collected, how it is stored, the purpose for which it will be used, and the right to withdraw at any time without affecting the child’s participation status.

Additional compliance measures include IRB-aligned digital consent mechanisms, demographic information stored in anonymised form, parent-verified participation as a prerequisite for any data collection, and a pseudonymisation-ready database schema to support controlled dataset release. All data flows are designed to support later pseudonymisation and restricted academic export under a controlled-access license.

3.16 Testing and Deployment Preparation

To validate the functionality and usability of the platform, we got our website and database verified from domain experts. We also conducted a small pilot study amongst our peers. This study enabled us to test the complete end-to-end workflow, including role-based access, task flows,

save/submit/retake logic, file uploads, and the accessibility suite. Feedback from this pilot informed interface refinements, error-handling improvements, and adjustments to scoring and task presentation.

The platform is now stable for larger-scale deployment soon with actual schools.

Chapter 4

Results

This chapter describes the outcomes achieved by the end of Semester 2. The primary engineering milestones are described first, followed by observed platform behaviour during expanded testing, and a summary of the current state of the dataset pipeline.

4.1 Platform Deployment and Accessibility

The platform is now live at dyslexia.flamenlp.in, accessible over HTTPS from any device with a modern browser. The EC2 instance ran continuously during the testing period with no unplanned downtime. The RDS instance handled all database operations, including concurrent task submissions from multiple test users, without integrity violations or performance degradation. Media files were uploaded to and retrieved from S3 reliably, with no data loss observed. The move to AWS removed the dependency on university server availability and gives the platform the independence it needs for large-scale school deployment.

The Android APK was installed and tested on three different Android device models. The platform loaded correctly, all task types were functional through the WebView interface, and audio recording via the device microphone worked as expected. Camera access through the APK also functioned correctly on all tested devices.

4.2 Performance Statistics and Feedback

Post-task statistics were computed correctly for all applicable task types during testing. Typing speed metrics were verified against manually timed control trials and found to be consistent. The star-rating feedback widget appeared consistently after task submission and the ratings were correctly stored in the feedback table. Optional comments were captured and stored without any character encoding issues.

4.3 Camera and Audio Recording Validation

Camera and microphone consent was tested across desktop browsers (Chrome, Firefox) and mobile devices. In all cases, the browser permission dialog appeared correctly before any recording commenced, and the student’s choice: grant both, grant audio only, or decline, was correctly recorded and respected. The live video preview appeared in the expected position during task attempts and did not interfere with the task content. Video and audio toggle buttons correctly paused and resumed the respective streams without interrupting the task session. Uploaded video files were correctly stored in S3 and retrievable via the backend reference path. In the audio-only scenario, the video file was not created and the database record correctly reflected the absence of a video stream.

4.4 Task Library improvement

Following NCERT aligned task curation, the platform now hosts between six and twelve tasks per class across all supported grade levels. All tasks were reviewed against Bloom’s Taxonomy and assigned a cognitive level by a domain specialist. The Listening Comprehension and Visual Aptitude tasks were integrated into the same task engine as the existing tasks, with correct save-submit-retake behaviour confirmed through testing.

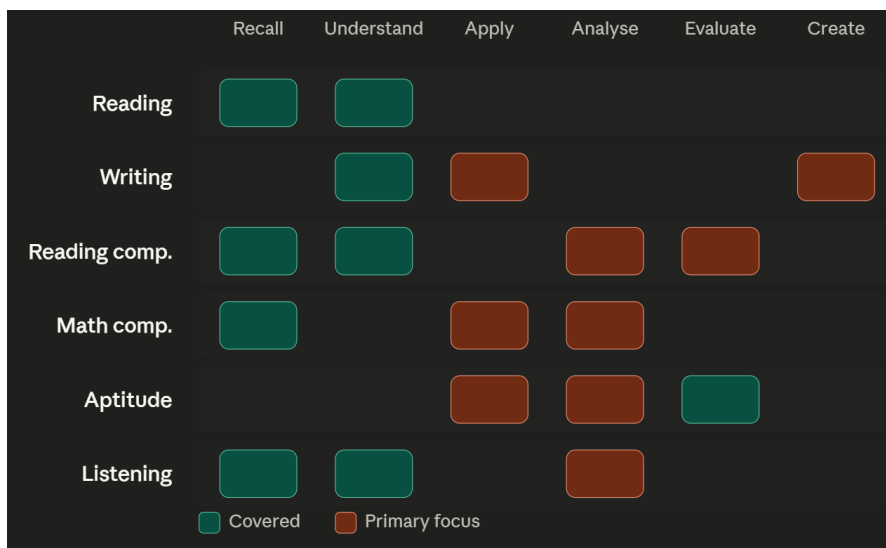


Figure 4.1: Task type × Bloom’s taxonomy matrix: maps all 6 task categories across Bloom’s 6 cognitive levels, showing which tasks cover which levels

4.5 UI and Responsiveness Validation

The redesigned landing and onboarding pages were reviewed with pilot users, and all reported a clearer understanding of the platform’s purpose compared to the Semester 1 version. The collapsible dashboard structure was consistently preferred by student testers for ease of navigation. No layout

failures were observed across the tested screen sizes or on the Android APK.

4.6 Summary

Semester 2 delivered a production-ready platform: publicly hosted, HTTPS-secured, legally documented, and significantly richer in data-capture capability than the Semester 1 prototype. The task library is more than doubled in size, two new cognitive task types have been added, and the platform now captures synchronized audio-video in addition to the multimodal behavioral signals available previously. The platform is ready for controlled deployment in partner schools and for the commencement of real-world data collection.

Chapter 5

Future Work

Future enhancements are planned to strengthen scalability, research usability, and linguistic diversity of the platform. The proposed developments include:

- **Real-World School Deployment and Data Collection** The immediate next step is onboarding partner schools in Delhi and beginning real-world data collection. The consent workflow and bulk onboarding tools are ready for this purpose.
- **Multilingual Task Expansion** The current task library is entirely in English. Hindi is the natural first target for expansion, given NCERT coverage, with regional languages (Tamil, Telugu, Kannada) planned subsequently. The schema already supports multilingual content.
- **SPL Educator Dashboard and Expert Annotation** The research utility of the dataset depends on the quality of its labels. A dedicated dashboard for Special Education practitioners is planned to enable expert annotation of audio recordings and typed responses, supporting production of research-grade gold labels with disagreement resolution.
- **iOS Application** iOS packaging requires an Apple Developer account and was not feasible this semester. It is planned for a future iteration pending funding.
- **Machine Learning Model Development** Once sufficient annotated data is collected, binary screening models will be trained using audio, keystroke, and comprehension timing features. Later stages will explore adaptive difficulty calibration and personalised intervention recommendations.
- **Difficulty-Level Surfacing and Adaptive Task Assignment** The Easy/Medium/Hard difficulty metadata will be curated from analysis of the data that we get, for adaptive assignment based on prior performance.
- **Dataset Release** The ultimate output of this project is a publicly available, ethically licensed dataset of multimodal dyslexia-related behavioral signals from Indian students. This dataset will be made available to the research community under a controlled-access license that requires researchers to agree to the terms of use, which will prohibit re-identification of

participants and restrict use to non-commercial academic research. The release is planned for after the first full wave of school deployment and annotation has been completed.

Chapter 6

Conclusion

This report described the work completed in the second semester of a B.Tech. project aimed at building an Indian-context multimodal dyslexia dataset collection platform. The Semester 1 report documented the complete technical foundation: a Flask backend, MySQL schema, role-based access control, six multimodal task types, an accessibility suite, and a gamified student interface. That foundation was built on a local server and validated through a small peer pilot.

This semester transformed that prototype into a production system. The most significant changes were the cloud migration to AWS (EC2, RDS, S3), the acquisition and configuration of a dedicated domain (dyslexia.flamenlp.in) with HTTPS encryption, and the legal finalization of the Terms of Service, Privacy Policy, and parental consent form. These three developments together address the practical requirements for real-world school deployment: the platform is reachable, secure, and legally defensible.

Beyond infrastructure, the semester delivered meaningful extensions to the platform’s data-capture capabilities. Synchronized audio-video recording with explicit per-student consent, controlled by real-time toggle buttons and a live preview display, adds a rich new behavioral channel. The task library was expanded from six tasks to six to twelve per class, with all content NCERT textbooks aligned and verified against Bloom’s Taxonomy by domain specialists. Two new task types: Listening Comprehension and Visual Aptitude — widen the cognitive signal space. Copy-paste prevention in writing tasks improves data quality. Post-task statistics give students immediate, engaging feedback. A star-rating feedback system provides operational intelligence for platform improvement. A parent profile page and an onboarding guide video reduce friction for new users. An Android APK makes the platform accessible on mobile devices.

Taken together, these contributions bring the project to the threshold of its central goal: the collection, at scale, of a multimodal dyslexia dataset representing Indian students. The infrastructure is in place, the legal framework is established, the task library reflects the actual curriculum these students follow, and the platform is deployed and accessible. The next phase, school-level deployment, annotation, and model development, now has a solid foundation to build on.

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